

THE TROTTER EXAM SOLUTIONS
ZIMSEC
MATHEMATICS 6042/1
2022 MARKING SCHEME BY
THE TROTTER 0774998145

ACKNOWLEDGMENTS: MR P JEKE (CASEMIRO)

#BE A GEM INSIDER

**I AM PROUDLY A GEM INSIDER,
MY FRIENDS ARE GEM INSIDERS,
MY CHILDREN ARE GEM INSIDERS
IT'S A GAME CHANGER
THIS RIGHT HERE IS THE FUTURE
EXAMS COMING, THERE IS NO DANGER**

GLOBETROTTER
EDUCATION MOVEMENT GEM
A realm of excellence

HIGH PASS RATE

ZIMSEC & Cambridge Experts

MAXIMUM COACHING BY RESULTS

MEET THE TROTTERS

Enroll at Globetrotter Education Movement Epworth Munyuki Centre for better results.
Form 1 to 6 enrollment in progress
Contact 0774998145 ...VISIT ALSO OUR YOUTUBE CHANNEL 'THE TROTTER ONLINE' FOR MATH VIDEOS.

This marking guide consists of 17 printed pages.

Hwatirera T The Trotter @GEM

[Turn Over]

WORKED AND TYPED BY TROTTER 0774998145

**VOICE OF
THE
PROFESSOR
(VOP)**

1. $(4 - 3x)^{-\frac{1}{2}}$

$$= 4^{-\frac{1}{2}} \left(1 - \frac{3x}{4}\right)^{-\frac{1}{2}}$$

$$= \frac{1}{4^{\frac{1}{2}}} \left[1 + \left(-\frac{1}{2}\right) \left(-\frac{3x}{4}\right) + \frac{\left(-\frac{1}{2}\right) \left(-\frac{1}{2} - 1\right) \left(-\frac{3x}{4}\right)^2}{2!} + \frac{\left(-\frac{1}{2}\right) \left(-\frac{1}{2} - 1\right) \left(-\frac{1}{2} - 2\right) \left(-\frac{3x}{4}\right)^3}{3!} \right]$$

$$= \frac{1}{2} \left[1 + \frac{3x}{8} + \frac{27x^2}{128} + \frac{135x^3}{1024} \right]$$

$$= \frac{1}{2} + \frac{3x}{16} + \frac{27x^2}{256} + \frac{135x^3}{2048}$$

Ans

(ii) $\left|\frac{3x}{4}\right| < 1$

$$= -1 < \frac{3x}{4} < 1$$

$$= -\frac{4}{3} < x < \frac{4}{3}$$

Ans

2. (i) $p = 0$

(ii) (a) $gh = \sqrt{h}$

$$\sqrt{x^3}$$

$$= (x^3)^{\left(\frac{1}{2}\right)}$$

$$= x^{\frac{3}{2}}$$

Ans

(b) $g^2 = gg = (\sqrt{x})^{\frac{1}{2}}$

$$= \left(x^{\frac{1}{2}}\right)^{\frac{1}{2}} = x^{\frac{1}{4}}$$

Ans

3. (i) $x = 3 + 4 \cos \theta$, $y = 1 + 4 \sin \theta$

$$\frac{x-3}{4} = \cos \theta$$

and

$$\frac{y-1}{4} = \sin \theta$$

$$\text{Using } \cos^2 \theta + \sin^2 \theta = 1$$

Award 4
marks

Award 1
mark

1 mark

Award 2
marks

Award 2
marks

$$\begin{aligned}\left(\frac{x-3}{4}\right)^2 + \left(\frac{y-1}{4}\right)^2 &= 1 \\ \frac{(x-3)^2}{16} + \frac{(y-1)^2}{16} &= 1 \\ (x-3)^2 + (y-1)^2 &= 16 \\ \text{which is an equation of a circle}\end{aligned}$$

Ans

**VOICE OF
THE
PROFESSOR
OR (VOP)**

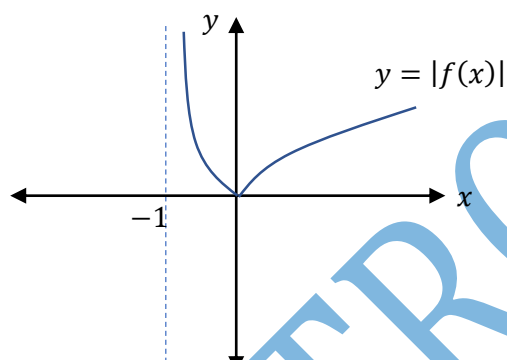
Award 3
marks

(ii) The centre is (3; 1)

$$\begin{aligned}\text{radius} &= \sqrt{16} \\ &= 4\end{aligned}$$

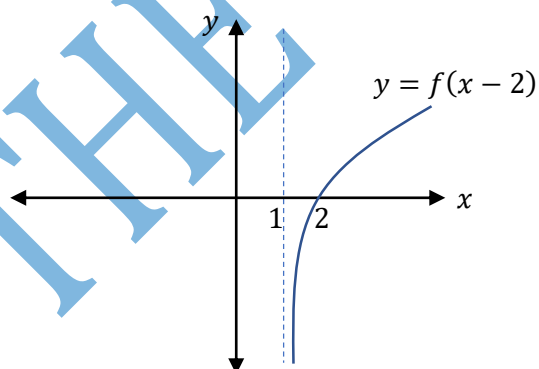
Award 2
marks

4. (a)

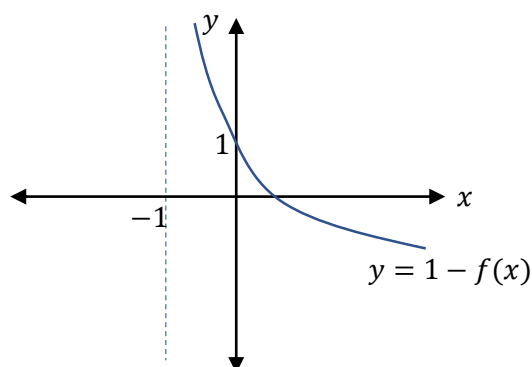


Award 2
marks

(b)



Award 2
marks

Award 2
marks

5. (a) Plot the graph of
- y
- against
- x^2

x^2	2.56	5.76	14.44	17.64	23.04	25.00
y	5.4	9.2	19.6	23.5	29.5	32.3

- (b)
- $b = y - \text{intercept} = 2.2$

$$a = \frac{y_1 - y_2}{x_1 - x_2}$$

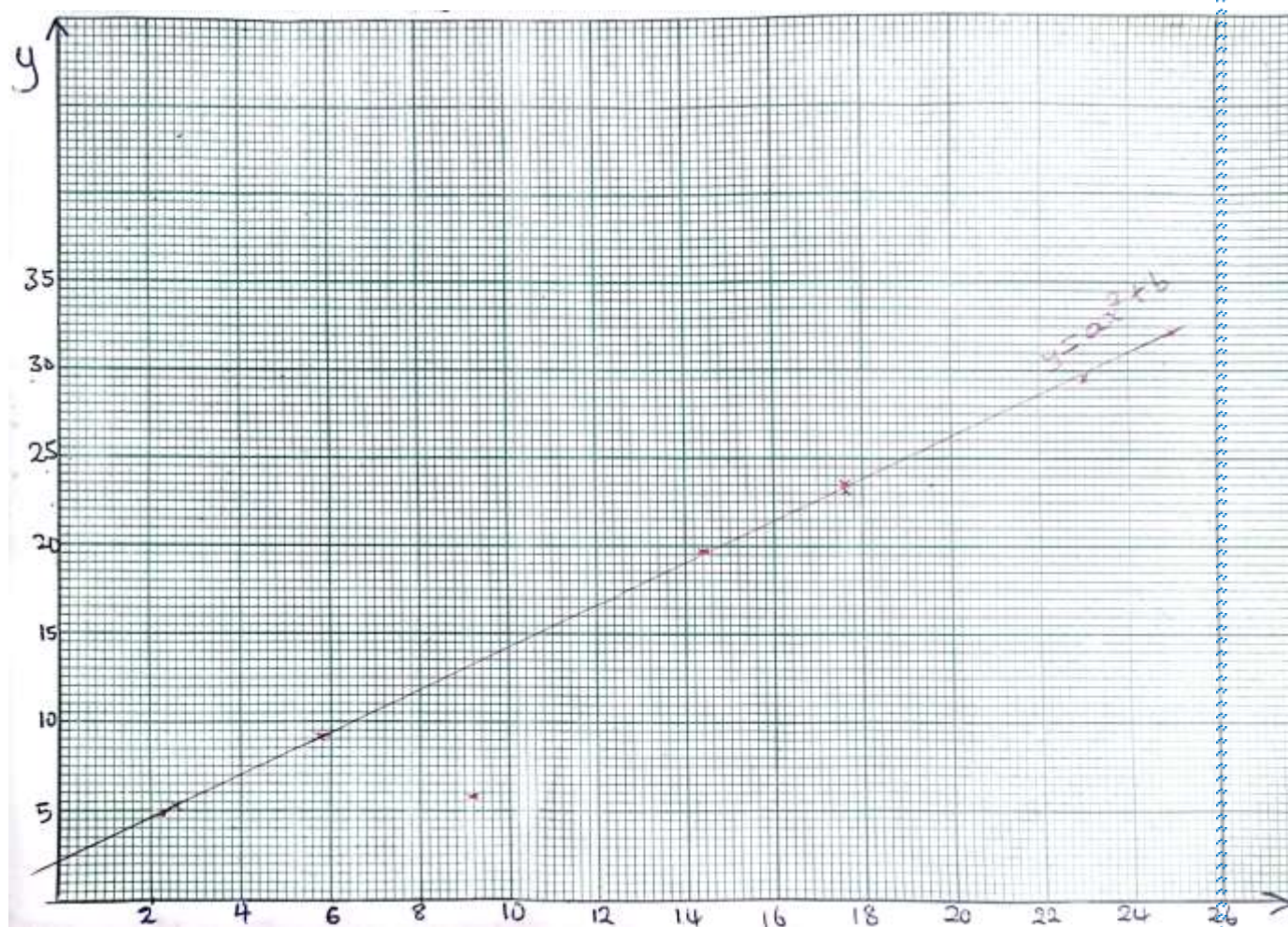
points to from the graph

 $(0; 2.2)$ and $(14; 19)$

$$= \frac{19 - 2.2}{14 - 0}$$

$$= 1.2$$

Award 3
marks



6. (a) $(x+1)^2 \frac{dy}{dx} + 2 = 0$

separating variables

$$(x+1)^2 dy = -2dx$$

$$dy = \frac{-2}{(x+1)^2} dx$$

$$\int dy = \int -2(x+1)^{-2} dx$$

$$y = \frac{2}{x+1} + c$$

when $x = 0, y = 1$

$$\therefore 1 = \frac{2}{0+1} + c$$

$$-1 = c$$

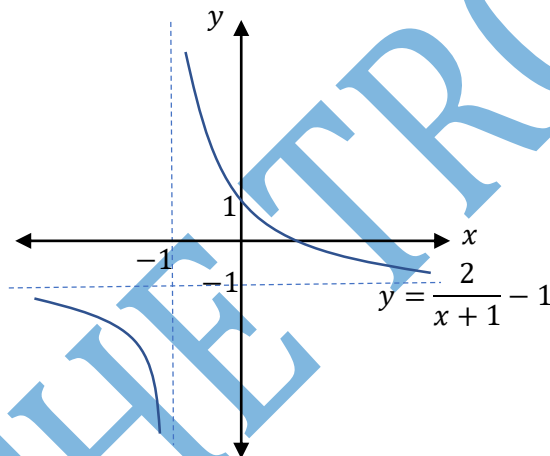
now

$$y = \frac{2}{x+1} - 1$$

$\therefore$ the particular solution is $y = \frac{2}{x+1} - 1$

Award 4
marks

(b)



Award 3
marks

7. (a) $C = hN + kD$

(b) $120 = 15h + 15k \dots (1)$

$90 = 10h + 20k \dots (2)$

$8 = h + k \dots (1)$

$9 = h + 2k \dots (2)$

$-1 = -k$

$1 = k$

$h = 7$

Award 3
marks

- (b) h represents the price of chicken
 k represents the price of a dozen of eggs

Award 1
 mark

(d) $C = hN + kD$

$$\begin{aligned} C &= 7N + D \\ \text{when } N &= 20, D = 30 \\ C &= 7(20) + 30 \\ &= \$170 \end{aligned}$$

Ans

Award 2
 marks

8.

$$\sum_{r=1}^n 8r - 1 = 4n^2 + 3n$$

when $n = 1$

$$\begin{aligned} LHS &= \sum_{r=1}^n 8r - 1 \\ &= 8(1) - 1 \\ &= 7 \end{aligned}$$

$$\begin{aligned} RHS &= 4(1)^2 + 3(1) \\ &= 4 + 3 \\ &= 7 \end{aligned}$$

$LHS = RHS$

Hence true for $n = 1$

Assume true for $n = k$

$$\sum_{r=1}^k 8r - 1 = 4k^2 + 3k$$

When $n = k + 1$

$$\sum_{r=1}^{k+1} 8r - 1 = 4(k+1)^2 + 3(k+1)$$

Proof:

$$LHS = \sum_{r=1}^{k+1} 8r - 1$$

$$\sum_{r=1}^k 8r - 1 + (k+1)^{th} \text{ term}$$

by the inductive hypothesis

$$= 4k^2 + 3k + 8(k + 1) - 1$$

$$= 4k^2 + 11k + 7$$

creating a situation

$$4k^2 + 8k + 4 + 3k + 3$$

$$= 4(k^2 + 2k + 1) + 3(k + 1)$$

$$= 4(k + 1)^2 + 3(k + 1)$$

$$\therefore LHS = RHS$$

Hence true for $n = k + 1$

Since the statement is true for n

$$= 1, n = k \text{ and } n = k + 1,$$

$\therefore$ by induction, it is true for $n \in \mathbb{N}$

Award 7
marks

9. (a) Let $f(x) = \frac{2}{4r^2 - 1}$

$$= \frac{2}{(2r - 1)(2r + 1)}$$

$$\therefore \frac{2}{(2r - 1)(2r + 1)} = \frac{A}{2r - 1} + \frac{B}{2r + 1}$$

$$2 = A(2r + 1) + B(2r - 1)$$

$$\text{let } r = \frac{1}{2}$$

$$2 = A(2)$$

$$1 = A$$

$$\text{let } r = -\frac{1}{2}$$

$$2 = B(-2)$$

$$-1 = B$$

Now

$$\frac{2}{4r^2 - 1} = \frac{1}{2r - 1} - \frac{1}{2r + 1}$$

Award 3
marks

(b)

$$\sum_{r=1}^n \frac{2}{4r^2 - 1} = \frac{2n}{2n + 1}$$

$$\text{consider LHS} = \sum_{r=1}^n \frac{2}{4r^2 - 1}$$

$$= \sum_{r=1}^n \frac{1}{2r - 1} - \frac{1}{2r + 1}$$

=

$$\begin{array}{r}
 \frac{1}{1} \quad - \quad \frac{1}{3} \\
 \frac{1}{3} \quad - \quad \frac{1}{5} \\
 \frac{1}{5} \quad - \quad \frac{1}{7} \\
 \vdots \\
 \vdots \\
 \vdots \\
 \frac{1}{2n-1} - \frac{1}{2n+1}
 \end{array}$$

$$\begin{aligned}
 &= 1 - \frac{1}{2n+1} \\
 &= \frac{2n+1-1}{2n+1} \\
 &= \frac{2n}{2n+1} \quad \text{As required}
 \end{aligned}$$

Award 6 marks

10. (a) $\int x(4-x)dx$

$$\begin{aligned}
 &= \int (4x - x^2)dx \\
 &= \frac{4x^2}{2} - \frac{x^3}{3} + c \\
 &= 2x^2 - \frac{x^3}{3} + c
 \end{aligned}$$

(b)

$$\begin{aligned}
 \text{Area} &= \int_0^4 x(4-x)dx \\
 &= \left[2x^2 - \frac{x^3}{3} \right]_0^4 \\
 &= \left[2(4)^2 - \frac{(4)^3}{3} \right] - \left[2(0)^2 - \frac{(0)^3}{3} \right] \\
 &= 32 - \frac{64}{3} \\
 &= \frac{32}{3} \text{ units}^2
 \end{aligned}$$

Award 2 marks

(c)

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^4 [x(4-x)]^2 dx \\
 &= \pi \int_0^4 (4x - x^2)^2 dx \\
 &= \pi \int_0^4 (16x^2 - 8x^3 + x^4) dx \\
 &= \pi \left[\frac{16x^3}{3} - \frac{8x^4}{4} + \frac{x^5}{5} \right]_0^4
 \end{aligned}$$

Award 2 marks

$$\begin{aligned}
 &= \pi \left[\frac{16(4)^3}{3} - \frac{8(4)^4}{4} + \frac{(4)^5}{5} \right] - \pi \left[\frac{16(0)^3}{3} - \frac{8(0)^4}{4} + \frac{(0)^5}{5} \right] \\
 &= \pi \left[\frac{512}{15} \right] - 0 \\
 &= \frac{512}{15} \pi \text{ units}^3
 \end{aligned}$$

Award 5 marks

11. $x^2 + (3k - 2)x + k + 2$

consider discriminant $b^2 - 4ac < 0$

$$(3k - 2)^2 - 4(1)(k + 2) < 0$$

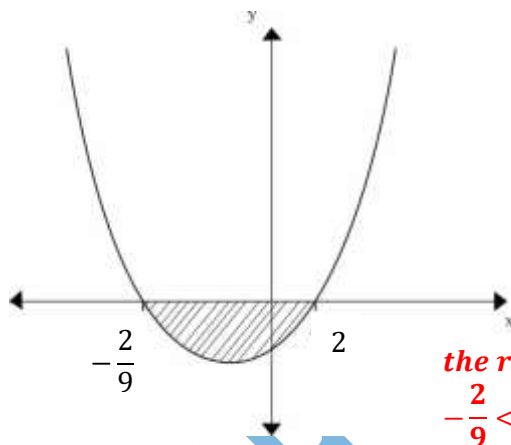
$$9k^2 - 12k + 4 - 4k - 8 < 0$$

$$9k^2 - 16k - 4 < 0$$

$$9k^2 - 18k + 2k - 4 < 0$$

$$9k(k - 2) + 2(k - 2) < 0$$

critical values are $k = -\frac{2}{9}$ or $k = 2$



the range is
 $-\frac{2}{9} < x < 2$

Award 5 marks

11 (b) $2x^3 - 9x^2 + 7x + 6 = 0$

finding the first factor using trial and error

$$f(2) = 0$$

$\therefore x - 2$ is a factor

Using long division

$$\begin{array}{r}
 2x^2 - 5x - 3 \\
 x - 2 \quad 2x^3 - 9x^2 + 7x + 6 \\
 \quad -(2x^3 - 4x^2) \\
 \quad \quad -5x^2 + 7x + 6 \\
 \quad \quad -(-5x^2 + 10x) \\
 \quad \quad \quad -3x + 6 \\
 \quad \quad \quad -(-3x + 6) \\
 \quad \quad \quad \quad 0
 \end{array}$$

solving the quotient

$$2x^2 - 5x - 3 = 0$$

$$2x^2 + x - 6x - 3 = 0$$

$$(2x + 1)(x - 3) = 0$$

$$x = 3 \text{ or } -\frac{1}{2}$$

$$\therefore \text{the roots are } x = 2 \text{ or } 3 \text{ or } -\frac{1}{2}$$

Award 5
marks

12. (a) $f(x) = ax^4 - 9x^3 + bx^2 - 3x + 2$

$$f(-2) = a(-2)^4 - 9(-2)^3 + b(-2)^2 - 3(-2) + 2 = 0$$

$$16a + 4b + 80 = 0$$

$$4a + b + 20 = 0$$

$$4a + b = -20 \dots (1)$$

$$f(1) = a(1)^4 - 9(1)^3 + b(1)^2 - 3(1) + 2 = -12$$

$$a - 9 + b - 1 = -12$$

$$a + b = -2 \dots (2)$$

solving simultaneously (1) - (2)

$$4a + b = -20 \dots (1)$$

$$a + b = -2 \dots (2)$$

$$3a = -18$$

$$a = -6 \text{ and } b = 4$$

Award 4
marks

(b)(i) $f(x) = -6x^4 - 9x^3 + 4x^2 - 3x + 2$

$$= -6\left(\frac{1}{2}\right)^4 - 9\left(\frac{1}{2}\right)^3 + 4\left(\frac{1}{2}\right)^2 - 3\left(\frac{1}{2}\right) + 2$$

$$= 0$$

Hence $1 - 2x$ is a factor (verified)

Award 2
marks

(ii) The quadratic factor is

$$= (x + 2)(1 - 2x)$$

$$= x - 2x^2 + 2 - 4x$$

$$= -2x^2 - 3x + 2$$

dividing (Use Long division)

$$\begin{array}{r} -6x^4 - 9x^3 + 4x^2 - 3x + 2 \\ -2x^2 - 3x + 2 \\ \hline \end{array}$$

we get the following quotient

$$3x^2 + 1$$

consider discriminant of the quotient

$$b^2 - 4ac$$

$$= (0)^2 - 4(3)(1)$$

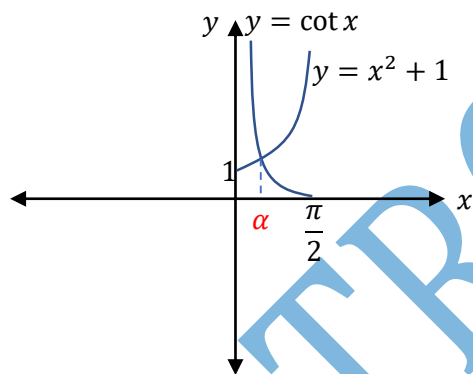
$$= -12$$

Since discriminant is < 0 , there are no real solutions from the quotient hence $f(x)$ has only 2 real roots

**VOICE OF
THE
PROFESSOR
(VOP)**

Award 4
marks

13. (a)



Award 3
marks

Since the graphs intersect only once the equation has one root

(b) Let $f(x) = \cot x - x^2 - 1$

$$f(0.6) = \cot 0.6 - 0.6^2 - 1$$

$$= 0.101695947 \dots + (ve)$$

$$f(0.9) = \cot 0.9 - 0.9^2 - 1$$

$$= -1.016448852 \dots - (ve)$$

Since there is a sign change a root lies between 0.6 and 0.9

Award 3
marks

(b)

$$X_{n+1} = \tan^{-1} \left(\frac{1}{1 + X_n^2} \right)$$

$$X_1 = \tan^{-1} \left(\frac{1}{1 + (06)^2} \right)$$

$$= 0.634022719$$

$$0.6340 \text{ (to 4dp)}$$

$$X_2 = 0.619579571$$

$$X_2 = 0.619579571$$

$$= 0.6196 \text{ (to 4dp)}$$

$$X_3 = 0.625737613$$

$$= 0.6257$$

Award 4
marks

$$14. A = \begin{pmatrix} 2 & 3 & -1 \\ 1 & 2 & 1 \\ 3 & 3 & 2 \end{pmatrix}$$

let C be the matrix of cofactors

$$C = \begin{bmatrix} + \begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} & - \begin{vmatrix} 1 & 1 \\ 3 & 2 \end{vmatrix} & + \begin{vmatrix} 1 & 2 \\ 3 & 3 \end{vmatrix} \\ - \begin{vmatrix} 3 & -1 \\ 3 & 2 \end{vmatrix} & + \begin{vmatrix} 2 & -1 \\ 3 & 2 \end{vmatrix} & - \begin{vmatrix} 2 & 3 \\ 3 & 3 \end{vmatrix} \\ + \begin{vmatrix} 3 & -1 \\ 2 & 1 \end{vmatrix} & - \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} & + \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} \end{bmatrix}$$

$$C = \begin{pmatrix} 1 & 1 & -3 \\ -9 & 7 & 3 \\ 5 & -3 & 1 \end{pmatrix}$$

$$\text{Det } A = 2(1) + 3(1) + (-1)(-3) \\ = 8$$

$$\text{Adjoint} = \begin{pmatrix} 1 & -9 & 5 \\ 1 & 7 & -3 \\ -3 & 3 & 1 \end{pmatrix}$$

$$A^{-1} = \frac{1}{8} \begin{pmatrix} 1 & -9 & 5 \\ 1 & 7 & -3 \\ -3 & 3 & 1 \end{pmatrix}$$

Award 7
marks

(b)

$$2x + 3y - z = 0$$

$$x + 2y + z = 4$$

$$3x + 3y + 2z = 4$$

$$\begin{pmatrix} 2 & 3 & -1 \\ 1 & 2 & 1 \\ 3 & 3 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix}$$

$$A^{-1} \begin{pmatrix} 2 & 3 & -1 \\ 1 & 2 & 1 \\ 3 & 3 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1} \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix}$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{8} \begin{pmatrix} 1 & -9 & 5 \\ 1 & 7 & -3 \\ -3 & 3 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{8} \begin{pmatrix} 2 \\ 2 \\ 10 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{1}{4} \\ \frac{1}{4} \\ \frac{5}{4} \end{pmatrix}$$

$$\therefore x = \frac{1}{4}, \quad y = \frac{1}{4}, \quad z = \frac{5}{4}$$

Award 4
marks

15. (a) (i) $OA = \begin{pmatrix} 3 \\ 3 \\ 4 \end{pmatrix} \quad OB = \begin{pmatrix} 2 \\ 5 \\ 3 \end{pmatrix} \quad OC = \begin{pmatrix} 3 \\ 4 \\ 3 \end{pmatrix}$

$$BA = OA - OB$$

$$= \begin{pmatrix} 3 \\ 3 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ 5 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

$$BC = OC - OB$$

$$= \begin{pmatrix} 3 \\ 4 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ 5 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

NOW

$$\cos ABC = \frac{BA \cdot BC}{|BA| \cdot |BC|}$$

$$\begin{aligned}\cos ABC &= \frac{\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}}{\sqrt{1^2 + (-2)^2 + 1^2} \cdot \sqrt{1^2 + (-1)^2 + 0^2}} \\ &= \frac{3}{\sqrt{12}} = \frac{\sqrt{3}}{2} \\ ABC &= \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) \\ &= 30^\circ\end{aligned}$$

Award 3 marks

a) (ii) $Area = \frac{1}{2} |BA \times BC|$

$$\begin{aligned}&= \frac{1}{2} \begin{vmatrix} i & j & k \\ 1 & -2 & 1 \\ 1 & -1 & 0 \end{vmatrix} \\ &= \frac{1}{2} \left| i \begin{vmatrix} -2 & 1 \\ -1 & 0 \end{vmatrix} - j \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} + k \begin{vmatrix} 1 & -2 \\ 1 & -1 \end{vmatrix} \right| \\ &= \frac{1}{2} |i + j + k| \\ &= \frac{1}{2} \sqrt{1^2 + 1^2 + 1^2} \\ &= \frac{\sqrt{3}}{2} \text{ units}^2\end{aligned}$$

Award 3 marks

Alternative solution

$$Area = \frac{1}{2} |BA| \cdot |BC| \sin \theta$$

we know from the previous stage that $\cos ABC = \frac{\sqrt{3}}{2}$

$$\therefore \sin ABC = \frac{1}{2} \quad \text{NB: sketch a triangle}$$

$$\text{We also have } |BA| \cdot |BC| = \sqrt{12} = 2\sqrt{3}$$

now

$$\begin{aligned}Area &= \frac{1}{2} (2\sqrt{3}) \left(\frac{1}{2}\right) \\ &= \frac{\sqrt{3}}{2} \text{ units}^2\end{aligned}$$

(iii) $AB = -BA$

$$\begin{aligned}
 & \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix} \\
 \mathbf{AF} &= \frac{1}{2} \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix} \\
 &= \begin{pmatrix} -\frac{1}{2} \\ 1 \\ -\frac{1}{2} \end{pmatrix} \\
 \mathbf{OF} &= \mathbf{OA} + \mathbf{AF} \\
 &= \begin{pmatrix} 3 \\ 3 \\ 4 \end{pmatrix} + \begin{pmatrix} -\frac{1}{2} \\ 1 \\ -\frac{1}{2} \end{pmatrix} \\
 &= \begin{pmatrix} \frac{5}{2} \\ 4 \\ \frac{7}{2} \end{pmatrix}
 \end{aligned}$$

Award 3
marks

15 (b) $CF = OF - OC$

$$\begin{aligned}
 & \begin{pmatrix} \frac{5}{2} \\ 4 \\ 7 \\ \frac{7}{2} \end{pmatrix} - \begin{pmatrix} 3 \\ 4 \\ 3 \end{pmatrix} \\
 &= \begin{pmatrix} -\frac{1}{2} \\ 0 \\ 1 \\ \frac{7}{2} \end{pmatrix}
 \end{aligned}$$

Consider dot product

 $CF \cdot AB$

$$\begin{aligned}
 & \begin{pmatrix} -\frac{1}{2} \\ 0 \\ 1 \\ \frac{7}{2} \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix} \\
 &= \frac{1}{2} + 0 - \frac{1}{2} \\
 &= 0
 \end{aligned}$$

since the dot product is = 0,
 CF is perpendicular to AB

Award 3
marks

Dear Reader

**CONSTRUCTIVE FEEDBACK ON THE LAYOUT AND ANY ERRORS AND OMISSIONS IS
ALWAYS WELCOME**

ENJOY YOUR STUDIES

BY T. HWATIRERA (THE TROTTER) 0774998145

thwatirera@gmail.com

‘Veni, Vidi, Vici... I came, I saw, I conquered’ *Gaius Julius Caesar*.

We sink black!

Siyaqeda umdlalo!

Tinobatanidza masports!

MATHEW 19:26 But Jesus looked at them and said, “With men this is impossible, but with God all things are possible.”